

VENTILATING SMALL CHICAGO HOMES

Because the affordable Chicago homes are airtight, builders were especially concerned with providing adequate ventilation and good indoor air quality.

**By ROBB ALDRICH and
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With the goal of creating healthy homes for people of modest means, Claretian Associates, a nonprofit community development organization, and South Chicago Workforce, a nonprofit builder and contractor, have teamed to create 26 unique homes in South Chicago, Illinois. These homes are constructed with structural insulated panels (SIPs), are heated with efficient condensing furnaces, utilize sealed-combustion water heaters, and contain efficient lights and appliances. The first 12 homes also feature 1.2 kW solar-electric systems. Claretian Associates' and South Chicago Workforce's commitment to high-quality affordable housing earned the project the city of Chicago's award for 2005 Outstanding Non-Profit Neighborhood Project.

Because the homes are so airtight (blower door tests showed 300–350 CFM₅₀), providing adequate ventilation is a special concern of the builder. David Sullivan of South Chicago Workforce and Paul Knight of Domus PLUS—an architect consulting on the project—approached Steven Winter Associates (SWA), the firm where the authors work, about monitoring the performance of several ventilation systems in the first three homes. SWA was able to perform the monitoring with funding from DOE's Building America program and from HUD's Partnership for Advancing Technology in Housing (PATH) program. The goal of the study was to evaluate the ventilation systems'



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performance with respect to fresh air distribution effectiveness and energy consumption and operating costs.

Testing Ventilation Strategies

In mid-October of 2004, monitoring systems and ventilation systems were completely installed and commissioned in the first three South Chicago homes. The three ventilation systems installed in the homes were

- house 1: energy recovery ventilation (ERV);

- house 2: Air Cyclor supply ventilation; and

- house 3: exhaust only (bath fans on timer control).

To examine ventilation effectiveness, SWA monitored

- temperature, humidity, and CO₂ concentrations outdoors, in two bedrooms, and in the living room (homes 1, 2, and 3);

- fresh-air supply flow rate, temperature, and humidity (homes 1 and 2); and

- exhaust air flow rate, temperature, and humidity (homes 1 and 3).

Table 1. Ranges in CO₂ Concentrations (ppm)

Month	1—ERV	2—Air Cycler	3—Bath Exhaust
Nov	124	80	160
Dec	120	85	83
Jan	132	102	79
Feb	128	85	88
Mar	150	93	88
Apr	185	77	164
Average	140	87	110

House 1

The first home completed features a RenewAire energy recovery ventilator (ERV) ducted into the central supply and return ducts in the basement. Before

home. CO₂ levels decreased markedly after this control was added. When ventilation systems were commissioned in October 2004, the outdoor air flow was measured at 80 CFM. ERV duty cycle was set to 75% to achieve the equivalent of 60 CFM continuous as required by ASHRAE 62.2.

The sensible effectiveness of the ERV was measured at 70%–75% during the winter months. The ERV draws 102 watts when operating; the air handler, which operates whenever the ERV operates, draws 712 watts in fan-only mode.

House 2

The second home features an Air Cycler supply ventilation system. The Air Cycler is a timer control that operates the central air handler at programmed intervals. A 6-inch duct runs from the return plenum to the outdoors, so some fresh air is introduced to the central air system and distributed through the heating ducts. A motorized damper is installed in the outside air duct which opens only when it receives a call for fresh air from the Air Cycler.

During commissioning in October 2004, outside air flow through the duct was measured at 80 CFM. The Air Cycler was programmed so that the central air handler would operate 75% of the time (15 minutes on, 5 minutes off) to achieve the 60 CFM effective continuous rate (per ASHRAE 62.2). The central air handler draws 712 watts when operating in fan-only mode to provide ventilation.

House 3

The third house contains two efficient bathroom exhaust fans (Broan model S80-UE) connected to two Tamarack Airetrack controllers. When operating, the first- and second-floor fans draw 56 and 59 CFM, respectively. The Airetrack controllers operate both fans at full speed for 35 minutes each hour to achieve the effective continuous rate of at least 60 CFM. Because the Airetrack controller can be programmed to operate only in 5-minute increments each hour, the effective continuous flow is slightly higher: 67 CFM. The Airetrack con-

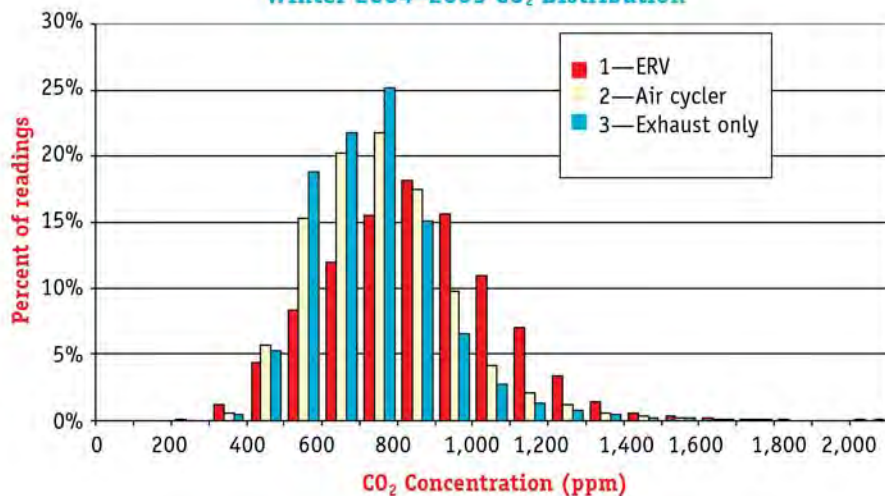
Winter 2004–2005 CO₂ Distribution

Figure 1. To evaluate the quality of fresh-air distribution, SWA has relied primarily on CO₂ concentration measurements. This histogram is for the entire 6-month test period.

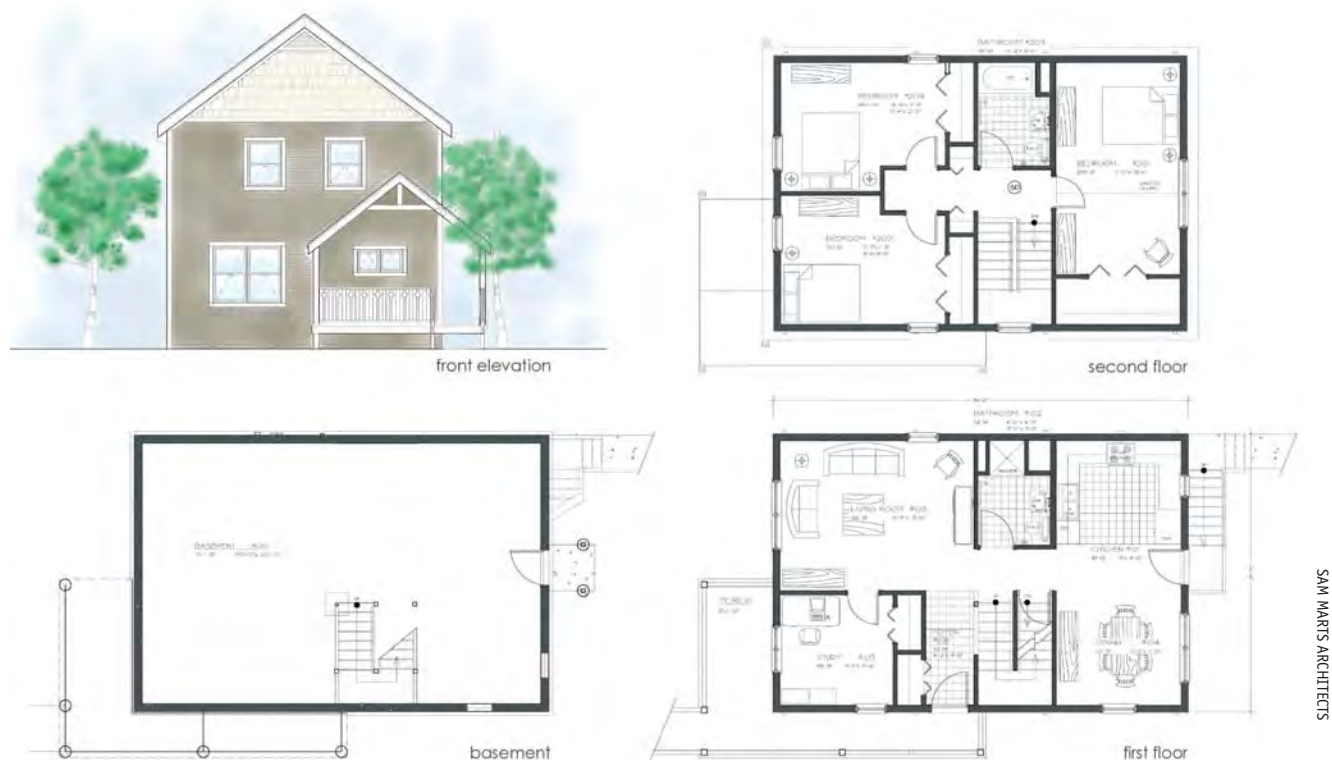


This RenewAire ERV was installed in one of the three South Chicago homes tested by SWA.

All sensors in the homes were connected to Campbell Scientific data logger systems. Readings were collected from sensors every ten seconds and recorded every 15 minutes. SWA finished collecting data at the end of April 2005; six full months of data were collected (November through April).

commissioning, the ERV operated constantly—24 hours a day—while the central air handler operated very infrequently (because of the low-heat load and city-mandated, oversized equipment).

Even with the ERV operating continuously, CO₂ levels were almost always above the 1,000 ppm ASHRAE-recommended maximum and often were above 2,000 ppm. Because the ERV makes use of the home's heating ducts, SWA surmised that most of the fresh air was being delivered to the basement or was simply short-circuiting through the duct system. To test this theory, SWA installed a controller that engages the central air handler fan whenever the ERV operates, thereby ensuring that fresh air is distributed throughout the



Here are house plans for the South Chicago affordable homes shown in the photo on p.24. Claretian Associates' and South Chicago Workforce's commitment to high-quality affordable housing earned the project the city of Chicago's award for 2005 Outstanding Non-Profit Neighborhood Project.

troller also allows occupants to turn the fan on at any time. Both fans operating together draw 45 watts.

Ventilation System Performance

SWA installed flow stations in both the exhaust ducts and the fresh-air ducts of the ERV system, as well as in the outside air duct in the Air Cyclor system. Flow rates of the exhaust fans were measured when the bath fans were installed with an Alnor Low-Flow balometer. Unfortunately, SWA was not able to install flow stations in the exhaust fan ducts; therefore these measured flow rates were assumed constant throughout the six-month test period. Although all ventilation systems were programmed at commissioning to provide 60 CFM of effective, continuous ventilation, monitoring shows that these rates were not maintained over time. Over time, the data show higher flow rates for the ERV system and lower flow rates for the Air Cyclor system.

Part of this flow rate discrepancy could have been caused by wind during commissioning, which might have

affected measured flow rates. In the Air Cyclor system, however, a steady decrease was observed in the first three months after commissioning. In the second half of October, effective continuous flow was 51 CFM. In November it fell to 44 CFM, and during the rest of the monitoring period, effective continuous flows were 32–36 CFM. SWA suspects that fouling of the intake screen is partly responsible for this decline. (These screens can get badly clogged, and maintenance is nonexistent.)

The higher flow rates in the ERV system cannot be explained in the same way. It is interesting to note a decrease (from 80 CFM to 70 CFM) in ERV flows during February. This decline was probably caused by homeowners adjusting the

controls of the ERV. Recorded ERV operating time was reduced from 70% to 60% during February; it is likely this was due to homeowner behavior. No such duty cycle change occurred in the Air Cyclor system.

Table 2. Energy Consumption for Ventilation During 6-Month Test Period

Ventilation System	Electricity kWh/Yr	Gas Therms/Yr	Total Ventilation Cost
ERV + AHU	1,834	-33	\$162
Air Cyclor	1,806	-14	\$171
Exhaust Only	114	97	\$ 75

Table 3. Energy Consumption for Ventilation Assuming Efficient 250W Furnace Fan During 6-Month Test Period

Ventilation System	Electricity kWh/Yr	Gas Therms/Yr	Total Ventilation Cost
ERV + AHU	829	3	\$82
Air Cyclor	634	28	\$82
Exhaust Only	114	97	\$75



(above) SWA studied ventilation strategies for small homes. This outside air intake duct features an Air Cyclor-controlled damper. (left) The South Chicago homes are constructed with structural insulated panels (SIPs), are heated with efficient condensing furnaces, utilize sealed-combustion water heaters, and contain efficient lights and appliances.

Because the homes are occupied by different people with different living patterns, however, an entirely straightforward comparison is not possible. Still, the tightest, lowest CO₂ concentration distribution indicates better fresh-air distribution. The histogram for the entire six-month period is shown in Figure 1, p.25. The histograms present all data points from all rooms at all times.

Another indicator of fresh-air distribution effectiveness is the range in CO₂ concentrations throughout each house at the same time. To clarify, the “range” in CO₂ concentrations discussed here is the maximum concentration in one home minus the mini-

num concentration at the same time in the same home. In homes with the best distribution, this CO₂ range will be small. See Table 1, p.25, which shows the average ranges during each month of the analysis.

- in the living room (central, first floor);
- in the master bedroom (second floor); and
- in the secondary bedroom (second floor).

To evaluate the effectiveness of the distribution, SWA generated histograms of CO₂ concentrations for each month for each ventilation system. The histogram of the best-performing system should have the smallest (tightest) distribution as well as low CO₂ values. The occupancy of these homes is very similar. The first home (ERV) houses two adults, four children, and one dog; the second and third homes each house two adults, three children, and one dog.

According to the histograms, the Air Cyclor and Exhaust systems have the tightest distributions with the lowest CO₂ concentration peaks. The data in Table 1 confirm this, showing that the Air Cyclor system has the least variability in CO₂ range, followed by the exhaust-only system. The ERV system has the highest variability. Because of the variability in occupant habits, however, SWA cannot definitively label any one system as “better” or “best.”

It is interesting that in data from the exhaust-only house, CO₂ ranges are higher in November and in April (see Table 1). During these months the furnace operation is lowest (20% and 8% run fractions, respectively). While some of this may be coincidental, it certainly suggests that air handler operation does indeed improve fresh air distribution.

Table 4. Annual Modeling Results with 712W Furnace Fan

Ventilation System	Electricity kWh/Yr	Gas Therms/Yr	Total Ventilation Cost
ERV + AHU	4,309	-51	\$398
Air Cyclor	3,714	12	\$379
Exhaust Only	263	76	\$ 76

Table 5. Annual Modeling Results with 250W Furnace Fan

Ventilation System	Electricity kWh/Yr	Gas Therms/Yr	Total Ventilation Cost
ERV + AHU	1,899	- 9	\$184
Air Cyclor	1,304	54	\$165
Exhaust Only	263	76	\$ 76
ERV w/o AHU	595	13	\$ 68

Carbon Dioxide and Fresh Air Distribution

To evaluate the quality of fresh-air distribution, SWA has relied primarily on CO₂ concentration measurements. A



PAUL KNIGHT



PAUL KNIGHT

SIPs were used in the South Chicago project to give homes tight, well-insulated envelopes. Blower door tests on the first homes showed leakage rates of only 300–350 CFM₅₀.

Ventilation Energy: Electricity

Knowing the actual power consumption and operation times of the ventilation equipment, it is fairly easy to assess how much electricity each ventilation system consumed in the six-month period. The ventilation electricity is the sum of the electrical energy consumed by the ventilation fans themselves and, where applicable, the electricity for the additional operation of the air handler (see Table 2, p.26).

Although the furnaces in these homes have high combustion efficiencies (92.5%), they have very poor electrical efficiencies (fan power consumption of 712 watts). This results in very high ventilation-related electric loads for the first two homes. The oversized furnaces exacerbate the problem—smaller furnaces would run much more frequently during the winter and much less operation would be required for ventilation alone.

Ventilation Energy: Natural Gas

Clearly, bringing outdoor air into the home during the winter increases the space-heating load. The ERV recovers 70%–75% of the energy in exhaust air, but the other two systems must rely on the furnace to heat the fresh air. Increased gas consumption caused by ventilation is shown in Table 2. A slight silver lining to the significant electricity consumption of the furnace fans is that this electric energy contributes to space heating. Because of this, gas consumption for the first two homes is actually

less than for a home with no ventilation at all. This reduction in gas consumption is evident in the homes' gas bills.

Ventilation Energy: Costs

The final column of Table 2 shows energy costs associated with ventilation. Cost for natural gas in South Chicago was \$0.65 per therm in the winter of 2004–2005; electricity cost \$0.10/kWh.

Efficient Air Handlers

The most striking component of ventilation energy consumption is the electricity used by the air handlers in the homes. It is unfortunate that efficient brushless permanent magnet motor (BPM) furnaces were not installed. While SWA has strongly recommended that more efficient furnace motors be used, these motors are typically available only on very expensive, high-end furnaces. In the South Chicago homes, the additional cost for this equipment would have been near \$1,000; the budget did not allow for this. SWA feels that it is important, however, to investigate the energy consumption of these ventilation systems with efficient air handlers. To do this, SWA repeated the energy calculations above using an air handler power consumption value of 250 watts in lieu of the measured 712 watts (see Table 3, p.26).

Modeling Annual Performance

SWA was able to obtain only six months of data for these homes (comprising most of the 2004–2005 heating sea-

son). To assess the operating costs for each ventilation system over an entire year, SWA calculated energy use of each system based on recorded 15-minute weather data from the 2004 calendar year. To more accurately compare the systems' relative performances, SWA also normalized the operating times and air flow rates in the modeling.

For the annual simulations, each ventilation system was modeled to deliver 90 CFM and to operate 67% of the time to meet the 60 CFM continuous equivalent required by ASHRAE 62.2. Table 4 shows results of this annual modeling using measured power consumption of the furnaces installed (712 watts). Table 5 shows the simulation results using an efficient air handler (250 watts). During the winter, the internal gains from additional furnace fan operation required by the ERV and the Air Cyclor largely offset additional gas loads for heating outdoor air. The negative numbers represent a reduction in net gas use caused by the ventilation system. There is no air conditioning in these modeled systems. If cooling were installed, internal gains from air handlers (when used for ventilation) would result in higher incremental ventilation costs.

The last line of Table 5 shows results of modeling an ERV system that does not use the central air handler to distribute air. Such a system requires separate ductwork to deliver ventilation air. The ERV in the first South Chicago home was originally ducted into the heating ducts in the basement without controls

to operate the furnace fan. This proved entirely unacceptable, as CO₂ concentrations were often above 2,000 ppm in the living spaces before SWA installed controls to distribute fresh air with the air handler. Running dedicated ventilation ducts certainly carries extra first costs, but, as shown above, it provides considerable energy and cost savings.

Best Choices

Among the systems monitored, it is clear that the exhaust-only system is the least costly to operate; costs were less than 50% of those of the other systems. Looking at CO₂ concentrations and air distribution, SWA believes the data show that the exhaust-only system does provide adequate distribution, and that it is the most appropriate system for these homes.

It is worth reiterating that the very high energy costs for the Air Cyclor and ERV systems would be significantly reduced with electrically efficient, properly sized furnaces, or especially (in the case of the ERV) with a dedicated duct system for ventilation.

This study certainly does not suggest that exhaust-only systems are appropriate for all applications. The small homes that we studied had sealed-combustion appliances and an efficient exhaust fan in the center of each floor (each floor being approximately 850 ft² in area). The effectiveness of exhaust strategies in larger homes requires more investigation, and the need for active air distribution may grow as house size increases. Nevertheless, this study shows that exhaust-only ventilation certainly can be a very effective, low-cost ventilation option.



Robb Aldrich and Gayathri Vijayakumar are engineers with Steven Winter Associates, Incorporated, in Norwalk, Connecticut. They focus on energy-efficient home design, evaluation, and research into advanced building practices and technologies.

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FOR MORE INFORMATION:

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